

A flexible and robust **network generation** pipeline for more detailed **bicycle infrastructure** **representations in MATSim**

AgiMo

Data-driven agile planning
for responsible mobility

SFB
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Berlin, new network: dedicated bicycle infrastructure in blue · Map data © OpenStreetMap contributors



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MATSim User Meeting 2026 · Paris, 28 September 2026

Why a new pipeline?

Better data

OSM cycling data has become much richer, down to the minor ways cyclists use, and LiDAR terrain models give elevation along each link. But OSM will never be fully consistent.

Robustness

So the pipeline has to cope with messy tagging: about 40 tags are evaluated, and different taggings of the same infrastructure end up in one category.

Consistency and applicability

Before: raw OSM tags. A measure meant editing tags.

Now: clear categories from a cyclist's perspective. A measure changes the category.



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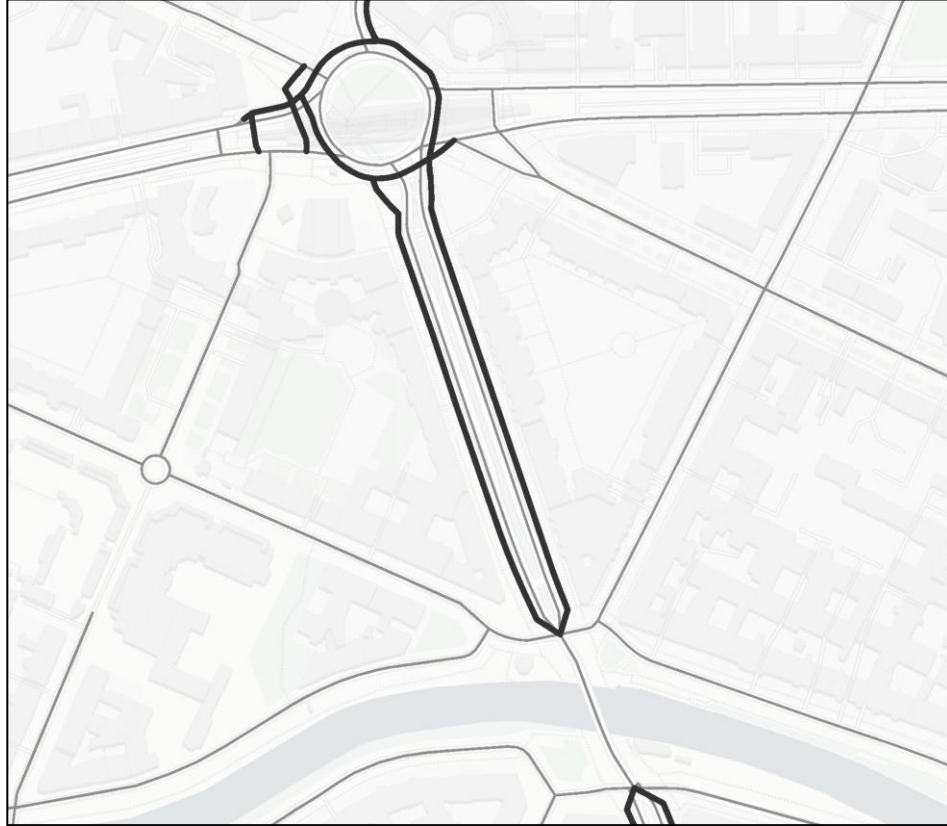
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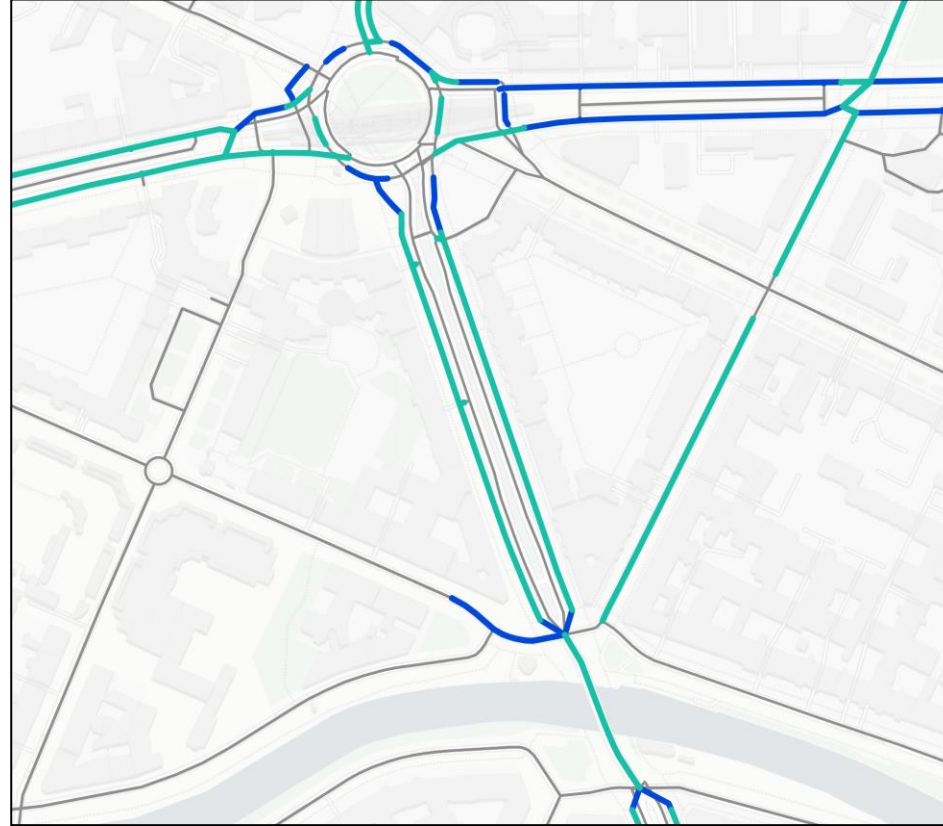
What the new network adds

Open Berlin 7.0 already marks where dedicated cycling infrastructure exists. The new network adds *which kind*, and the minor ways in between.



(a) MATSim Open Berlin Scenario 7.0

— links — dedicated cycling infrastructure



(b) New network

— links — on the road — separate cycleway

Kottbusser Tor, Berlin-Kreuzberg; (b) drawn from the network of 6 Aug 2026. Gray includes crossings and unclear cases.
Map data © OpenStreetMap contributors.

What the new network adds

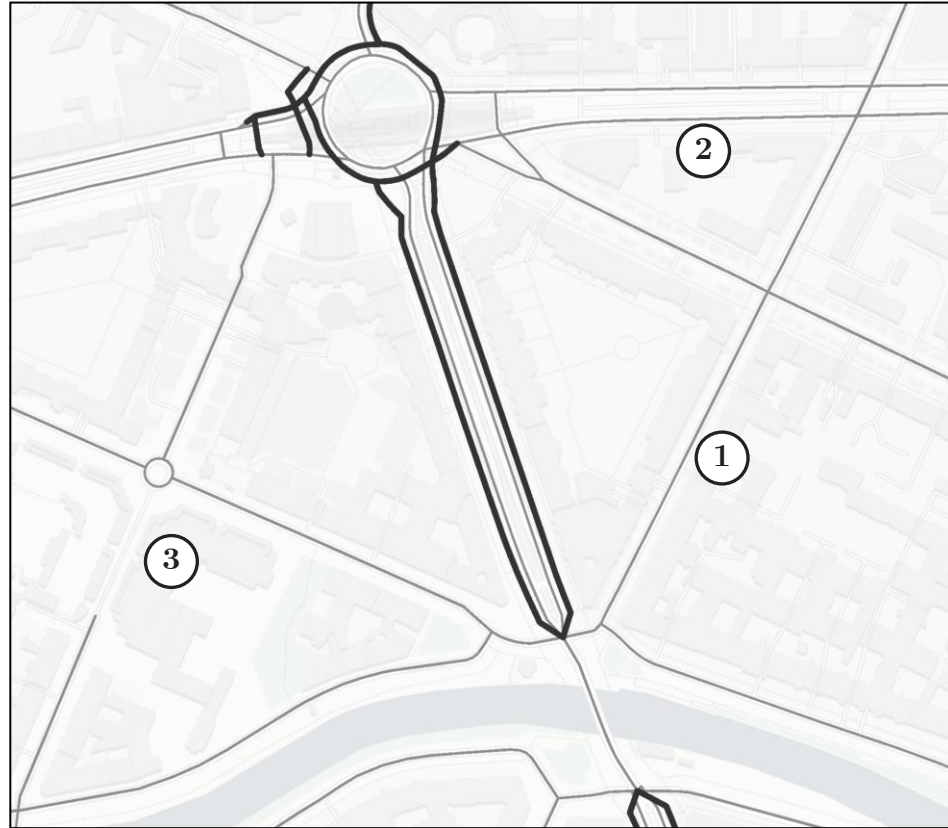


Kottbusser Tor on
radinfra.de

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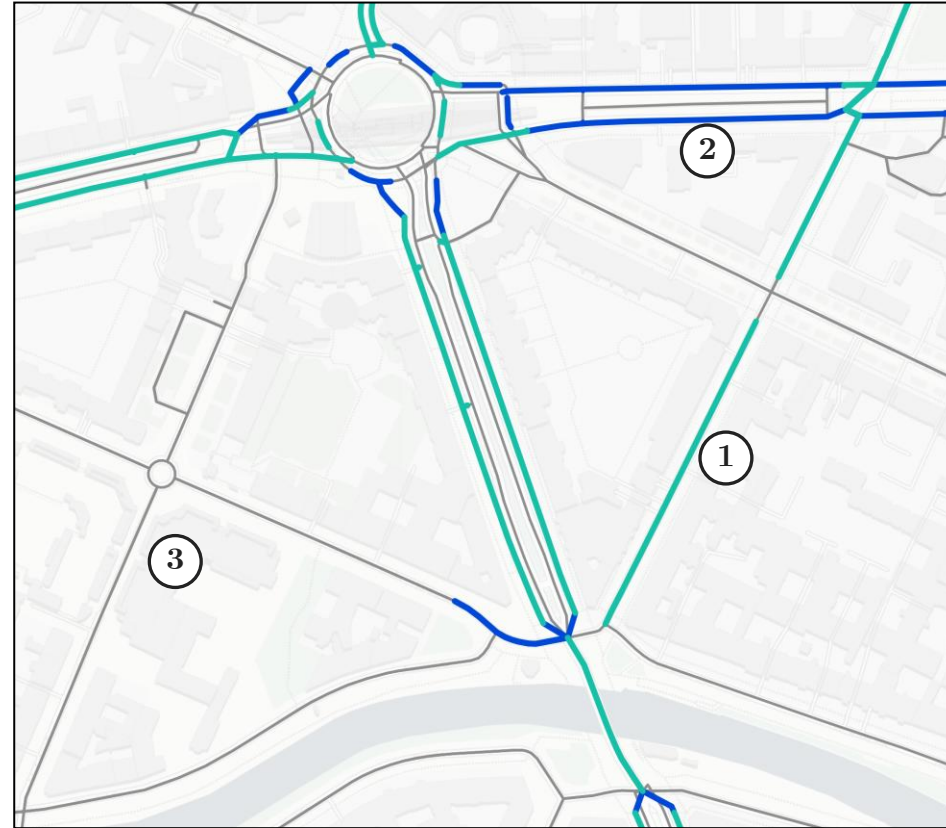
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- ① bicycle road ② cycle tracks along the road ③ a gap closed by a minor way



(a) MATSim Open Berlin Scenario 7.0

— links — dedicated cycling infrastructure



(b) New network

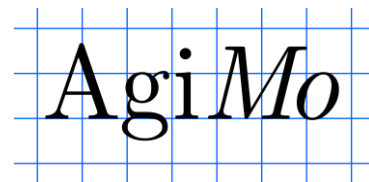
— links — on the road — separate cycleway

What the network has to provide

Every measure is compared against the baseline. So the network has to show today's cycling infrastructure as well as possible.

The tools add to a MATSim network:

- **cyclable minor ways:** footways, service roads, paths and tracks open to cycling
- the type of bicycle **infrastructure**, in 27 categories
- smoothness and surface as a fallback for the **comfort** component
- **elevation** along the actual link course



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Two ways in, same classification

Preparation

prepare extract
(osmosis)

.osm.pbf → clip to region,
mark the detailed area

Path 1: SUMO-netconvert

netconvert

2nd pass with a
keep-edges list from
bicycle-keep-edges

network-from-sumo

--keep-cyclable-minor-ways

bicycle-attributes

--dem --country
[--simplify]

Path 2: SupersonicOsmNetworkReader

bicycle-network --input --dem --dem-crs --country

classification during import · bicycle-aware simplification

Both paths

cleanup, split bike links

Same classification and
link attributes. The
networks themselves
differ: junction joining,
capacities and freespeeds
follow each path's
defaults.

Why two paths? In AgiMo the tools have to fit into existing scenarios: Berlin, Munich and Dresden build their networks with SUMO-netconvert (path 1). Path 2 builds a network directly from OSM, in one command.

One classifier, 27 categories

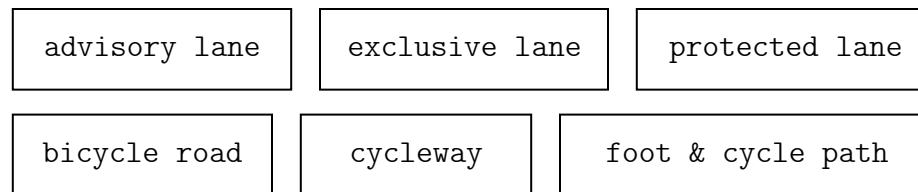
~40

relevant OSM tags per way: highway, the
cycleway:* variants, bicycle, traffic_sign,
access ...



27

cycling-infrastructure categories, stored as
one attribute per link



... and 21 more

The categories describe what a cyclist finds on the
ground: a lane on the road, a separate cycleway, a
path shared with people walking.

The category system is adopted from radinfra.de /
TILDA (FixMyCity, OSM Verkehrswende
community), developed over several years and
reviewed by the OSM community.

Traffic-sign rules per country: de, at (experimental), generic.

Robust to messy tagging

Same cycleway, mapped in two ways

highway=cycleway

is_sidepath=yes

→ CYCLEWAY_ADJOINING

highway=secondary

cycleway:right=track

→ CYCLEWAY_ADJOINING

Same rather incomplete tags, but different traffic sign

highway=path

segregated=yes

traffic_sign=DE:237

→ CYCLEWAY_ADJOINING

highway=path

segregated=yes

traffic_sign=DE:241

→ FOOT_AND_CYCLEWAY_
SEGREGATED_ADJOINING



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27 categories, combined as needed



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On the road

lanes on the carriageway 533 km
CYCLEWAY_ON_HIGHWAY_ADVISORY
CYCLEWAY_ON_HIGHWAY_EXCLUSIVE
CYCLEWAY_ON_HIGHWAY_ADVISORY_OR_EXCLUSIVE
CYCLEWAY_ON_HIGHWAY_PROTECTED
CYCLEWAY_ON_HIGHWAY_BETWEEN_LANES

bicycle roads 103 km
BICYCLE_ROAD
BICYCLE_ROAD_VEHICLE_DESTINATION

bus lanes & sharrows 69 km
SHARED_BUS_LANE_BUS_WITH_BIKE
SHARED_BUS_LANE_BIKE_WITH_BUS
SHARED_MOTOR_VEHICLE_LANE

Separate cycleways

cycleways 1,086 km
CYCLEWAY_ADJOINING
CYCLEWAY_ISOLATED
CYCLEWAY_ADJOINING_OR_ISOLATED
CYCLEWAY_LINK

Not colored

crossings, unclear cases 218 km
CROSSING
NEEDS_CLARIFICATION

no infrastructure 11,943 km
NONE

Shared with pedestrians

footways with cycling allowed 1,009 km
FOOTWAY_BICYCLE_YES_ADJOINING
FOOTWAY_BICYCLE_YES_ISOLATED
FOOTWAY_BICYCLE_YES_ADJOINING_OR_ISOLATED
PEDESTRIAN_AREA_BICYCLE_YES

foot & cycle paths 745 km
FOOT_AND_CYCLEWAY_SHARED_ADJOINING
FOOT_AND_CYCLEWAY_SHARED_ISOLATED
FOOT_AND_CYCLEWAY_SHARED_ADJOINING_OR_ISOLATED
FOOT_AND_CYCLEWAY_SEGREGATED_ADJOINING
FOOT_AND_CYCLEWAY_SEGREGATED_ISOLATED
FOOT_AND_CYCLEWAY_SEGREGATED_ADJOINING_OR_ISOLATED

The network stores the 27 values; the grouping is up to the use. Three groups for a map, a few classes for a simple scoring, all 27 for a detailed one.

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A link in the new network

```
<link id="1435070341#0" from="cluster_1135..." to="29276743"
  length="87.94" freespeed="5.831" capacity="600.0" permlanes="1.0"
  modes="bike,car,freight,ride,truck">
  <attributes>
    <attribute name="bicycle_infra">BICYCLE_ROAD_VEHICLE_DESTINATION</attribute>
    <attribute name="osm:bicycle">designated</attribute>
    <attribute name="osm:surface">asphalt</attribute>
    <attribute name="osm:smoothness">good</attribute>
    <attribute name="gradient">-0.001</attribute>
    <attribute name="max_gradient">-0.001</attribute>
    <attribute name="elevation_gain">0.0</attribute>
    <attribute name="elevation_loss">0.1</attribute>
  </attributes>
</link>
```

Mariannenstraße, the bicycle road (1) on the Kotti map (network of 6 Aug 2026)

Scoring can weight the categories directly instead of interpreting raw OSM tags.

Measures become
attribute edits:

New bicycle road

NONE → BICYCLE_ROAD

Protected lane

..._ADVISORY → ..._PROTECTED

The logo for AgiMo, featuring the text "AgiMo" in a serif font, with the "i" in "Agi" and the "o" in "Mo" having dots. The text is overlaid on a blue grid pattern.

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Bike links split off the carriageway

Before: one link, one queue



```
4711  modes="car,bike"  
bicycle_infra=CYCLEWAY_ON_HIGHWAY_EXCLUSIVE
```

After: a separate bike link

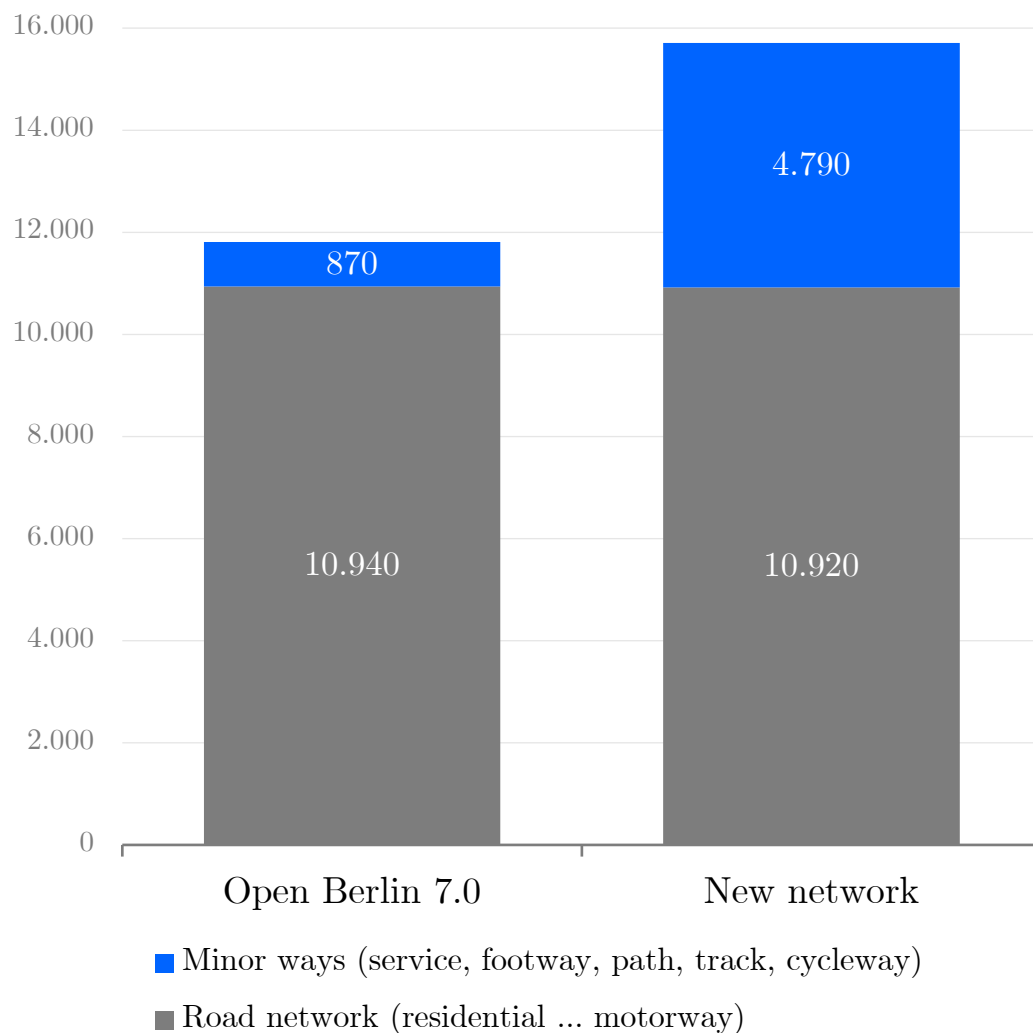


```
4711      modes="car"  
4711_bike modes="bike" bicycle_infra=...
```

- Where a lane or track is tagged on the road, bikes share one queue with cars. The PassingVehicleQ reorders free flow, but it does not dissolve a jam.
- The pipeline splits off a bike-only link with the same nodes and length. Category, surface and elevation move to the bike link.
- Bicycle roads, sharrows and pedestrian areas stay shared.



Berlin: road network and minor ways



+3,900 km

of minor ways **within the city limits:**
870 → 4,790 link-km

Footways and paths open to cycling, cycleways, service roads and tracks. They close gaps, even where they carry no dedicated bicycle infrastructure.

The number of links roughly doubles: 84,000 → 173,000.

Berlin: infrastructure categories

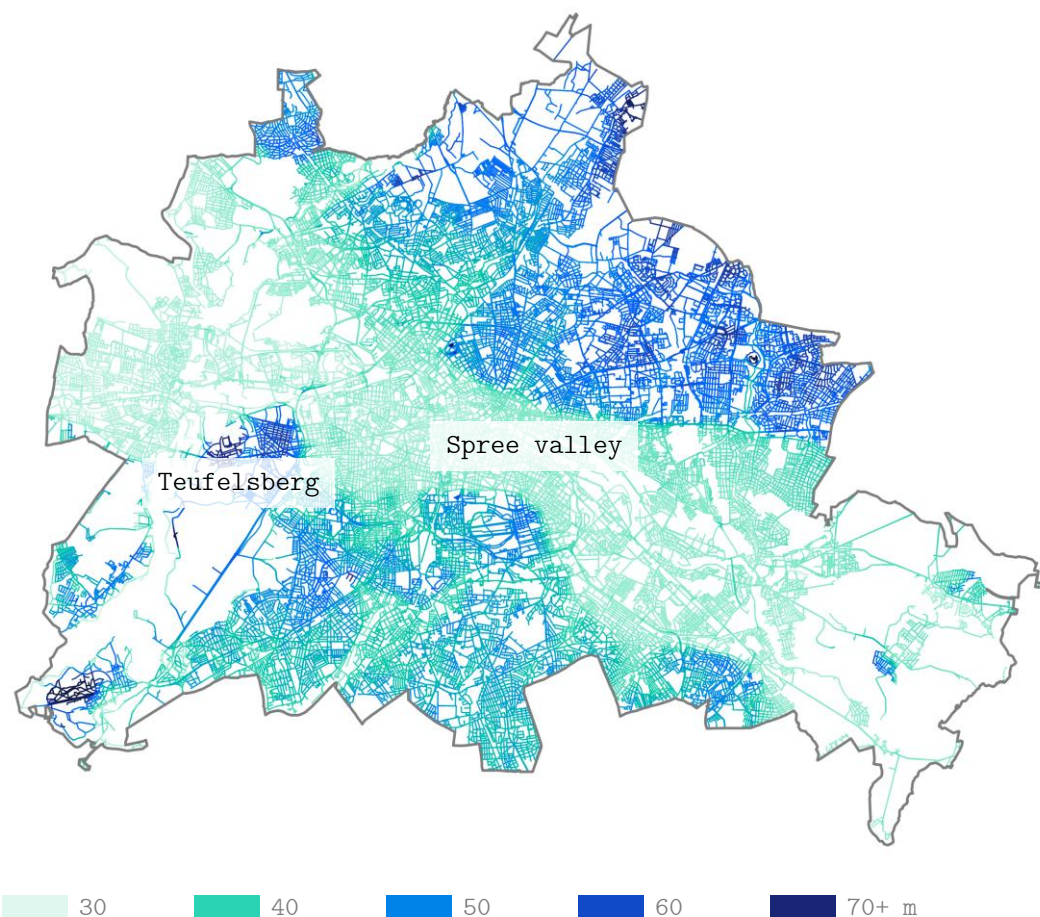


23%

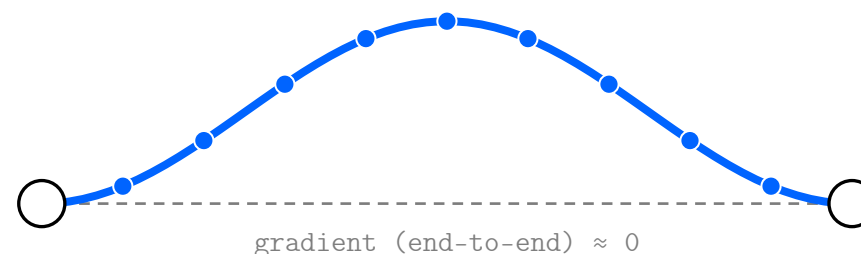
of the link-km are cycling infrastructure:
3,544 km

On the road	704 km
Separate cycleways	1,086 km
Shared with pedestrians	1,754 km
rest of the network	

Elevation along every link



$$\text{max_gradient} \cdot \text{elevation_gain} \cdot \text{elevation_loss}$$



Elevation is sampled from a LiDAR terrain model along the link geometry, not only at the two end nodes.

This follows route-choice studies, which describe hills by the maximum gradient, the climb per distance or the share of steep sections rather than the end-to-end gradient (Menghini et al. 2010; Hood et al. 2011; Broach et al. 2012).

What comes next

Validate

Run scenarios on the new networks and check capacities, freespeeds and runtime.

Categories in scoring

Pluggable scoring and speed parameters in the bicycle contrib, with an infrastructure factor per category. Existing setups keep working.

Into matsim-lib

Still work in progress, on a branch with a draft PR. Once validated, the tools move into the bicycle contrib.

In short: one infrastructure category per link, robust to messy tagging, plus minor ways and elevation, from SUMO-netconvert or straight from OSM.

Thank you!



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www.agimo-trr408.de



Meet me at my poster
Mapping missing cycling
infrastructure in OSM
from street-level imagery

hEART 2026, session 17E
Wed, 30 Sep, 17:15-18:45
Salle 116

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